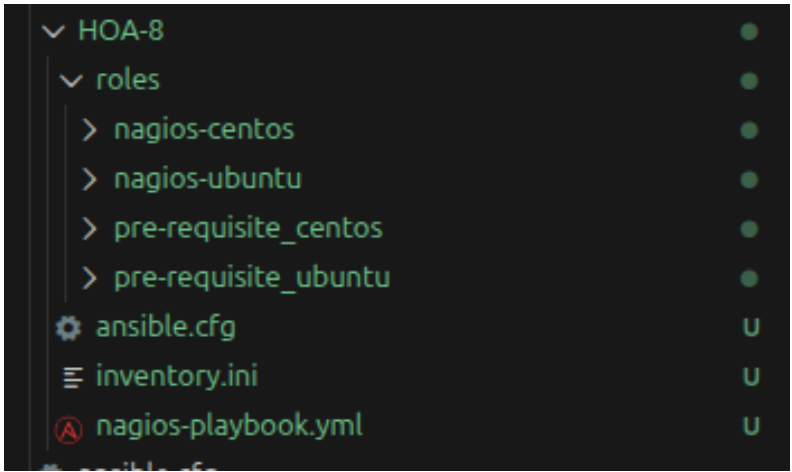


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Activity 8: Install, Configure, and Manage Availability Monitoring tools	
1. Objectives	
Create and design a workflow that installs, configure and manage enterprise monitoring tools using Ansible as an Infrastructure as Code (IaC) tool.	
2. Discussion	
Availability monitoring is a type of monitoring tool that we use if the certain workload is up or reachable on our end. Site downtime can lead to loss of revenue, reputational damage and severe distress. Availability monitoring prevents adverse situations by checking the uptime of infrastructure components such as servers and apps and notifying the webmaster of problems before they impact on business.	
3. Tasks	
1. Create a playbook that installs Nagios in both Ubuntu and CentOS. Apply the concept of creating roles. 2. Describe how you did step 1. (Provide screenshots and explanations in your report. Make your report detailed such that it will look like a manual.) 3. Show an output of the installed Nagios for both Ubuntu and CentOS. 4. Make sure to create a new repository in GitHub for this activity.	
4. Output (screenshots and explanations)	
 This shows all the yml files and roles that are used for the activity.	

The following are what contains inside the yaml files:

pre-requisite_ubuntu

```
Go ... ← → CPE212_Santiago
! main.yml M X
HOA-8 > roles > pre-requisite_ubuntu > tasks > ! main.yml
1 ---
2 - name: Install Nagios Pre-Requisites on Ubuntu
3   apt:
4     name:
5       - apache2
6       - php
7       - php-gd
8       - libapache2-mod-php
9       - unzip
10      - wget
11      - build-essential
12      - nagios-plugins
13    state: present
14    update_cache: yes
15    when: ansible_distribution == "Ubuntu"
16
17 - name: Enable Apache service
18   service:
19     name: apache2
20     state: started
21     enabled: yes
22    when: ansible_distribution == "Ubuntu"
23
24 - name: Enable Apache modules needed for Nagios
25   command: a2enmod rewrite
26   when: ansible_distribution == "Ubuntu"
27
28 - name: Restart Apache service
29   service:
30     name: apache2
31     state: restarted
32    when: ansible_distribution == "Ubuntu"
```

pre-requisite_centos

```
Go ... ← → CPE212_Santiago
! main.yml M X
HOA-8 > roles > pre-requisite_centos > tasks > ! main.yml
1 ---
2 - name: Install EPEL repository
3   dnf:
4     name: epel-release
5     state: present
6     when: ansible_distribution == "CentOS"
7
8 - name: Install Nagios Pre-Requisites on CentOS
9   dnf:
10    name:
11      - httpd
12      - php
13      - php-gd
14      - gcc
15      - glibc
16      - glibc-common
17      - wget
18      - unzip
19      - make
20      - net-snmp
21      - nagios-plugins
22    state: present
23    update_cache: yes
24    when: ansible_distribution == "CentOS"
25
26 - name: Start Apache on CentOS
27   service:
28     name: httpd
29     state: started
30     enabled: yes
31    when: ansible_distribution == "CentOS"
```

These yamls are tasked to install all dependencies for both Ubuntu and CentOS to install Nagios properly on both servers. EPEL repository was enabled to provide additional packages that are not

included in the default repositories. This repository contains the Nagios monitoring plugins required for the Nagios service to function properly.

nagios-centos

```
Go ... ← → CPE212_Santiago
! main.yml M X
HOA-8 > roles > nagios-centos > tasks > ! main.yml
1 ---
2 - name: Install Nagios on CentOS
3   yum:
4     name: nagios
5     state: present
6     when: ansible_distribution == "CentOS"
7
8 - name: Start Nagios Service on CentOS
9   service:
10    name: nagios
11    state: started
12    enabled: yes
13    when: ansible_distribution == "CentOS"
```

nagios-ubuntu

```
Go ... ← → CPE212_Santiago
! main.yml M X
HOA-8 > roles > nagios-ubuntu > tasks > ! main.yml
1 ---
2 - name: Install Nagios on Ubuntu
3   apt:
4     name: nagios4
5     state: present
6     update_cache: yes
7     when: ansible_distribution == "Ubuntu"
8
9 - name: Start Nagios Service on Ubuntu
10  service:
11    name: nagios4
12    state: started
13    enabled: yes
14    when: ansible_distribution == "Ubuntu"
```

Both nagios-centos and nagios-ubuntu are tasked to install Nagios and start the service.

nagios-playbook.yml

```
Go ... ← → CPE212_Santiago
! nagios-playbook.yml M X
HOA-8 > ! nagios-playbook.yml
1 ---
2 - name: Install and Start Nagios on Ubuntu & CentOS
3   hosts: web_servers
4   become: yes
5
6   roles:
7     - pre-requisite_ubuntu
8     - pre-requisite_centos
9     - nagios-ubuntu
10    - nagios-centos
```

This is the main playbook where I put all the roles needed for Installing and Starting Nagios on both Ubuntu and CentOS servers. Roles are implemented to ensure the organized and structured playbook.

OUTPUTS

The following screenshots are the outputs after running the ansible playbook.

```
TASK [pre-requisite_ubuntu : Install Nagios Pre-Requisites on Ubuntu] *****
skipping: [server4]

changed: [server1]

TASK [pre-requisite_ubuntu : Enable Apache service] *****
skipping: [server4]
ok: [server1]

TASK [pre-requisite_ubuntu : Enable Apache modules needed for Nagios] *****
skipping: [server4]
changed: [server1]

TASK [pre-requisite_ubuntu : Restart Apache service] *****
skipping: [server4]
changed: [server1]
```

```
TASK [nagios-ubuntu : Install Nagios on Ubuntu] *****
changed: [server1]

TASK [nagios-ubuntu : Start Nagios Service on Ubuntu] *****
changed: [server1]
```

This shows that the dependencies and the Nagios were successfully installed for Ubuntu server (server1). It also shows that the Nagios service has already started.

```
TASK [pre-requisite_centos : Install EPEL repository] *****
skipping: [server1]
changed: [server4]

TASK [pre-requisite_centos : Install Nagios Pre-Requisites on CentOS] *****
skipping: [server1]
changed: [server4]

TASK [pre-requisite_centos : Start Apache on CentOS] *****
skipping: [server1]
changed: [server4]
```

```
TASK [nagios-ubuntu : Install Nagios on Ubuntu] *****
skipping: [server4]
ok: [server1]

TASK [nagios-ubuntu : Start Nagios Service on Ubuntu] *****
skipping: [server4]
ok: [server1]

TASK [nagios-centos : Install Nagios on CentOS] *****
skipping: [server1]
changed: [server4]

TASK [nagios-centos : Start Nagios Service on CentOS] *****
skipping: [server1]
changed: [server4]

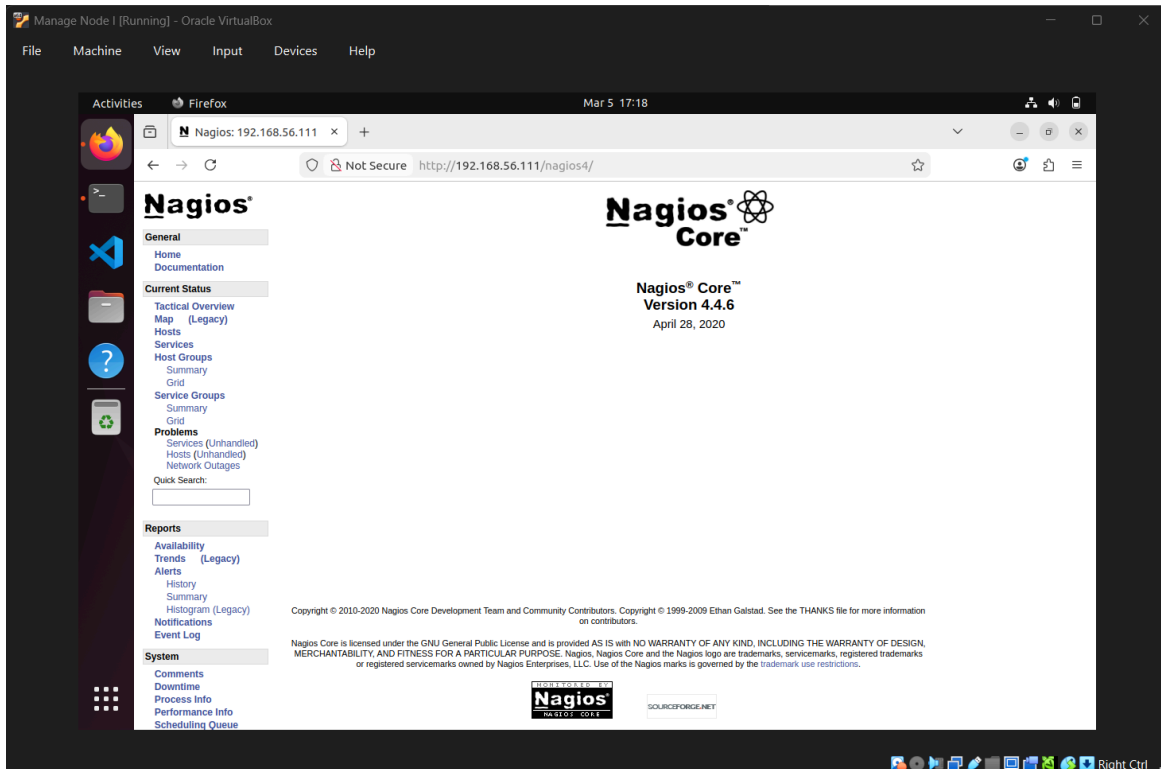
PLAY RECAP *****
server1      : ok=7    changed=2    unreachable=0    failed=0    skipped=5    rescued=0    i
gnored=0
server4      : ok=6    changed=5    unreachable=0    failed=0    skipped=6    rescued=0    i
```

This shows that the dependencies and the Nagios were successfully installed for CentOS server (server4). It also shows that the Nagios service has already started.

Verification

The following screenshots show the verification for Nagios installation and the service.

Ubuntu Server



This verifies the Nagios installation and the service has been started in Ubuntu. To verify it, I simply typed the IP address of the managed node and /nagios. This is the output.

CentOS Server

```
jalene@server4:~ — sudo systemctl status nagios
● nagios.service - Nagios Core 4.4.14
   Loaded: loaded (/usr/lib/systemd/system/nagios.service; enabled; preset: d>
   Active: active (running) since Thu 2026-03-05 17:41:24 PST; 1min 19s ago
     Docs: https://www.nagios.org/documentation
   Process: 906 ExecStartPre=/usr/sbin/nagios -v /etc/nagios/nagios.cfg (code=>
   Process: 918 ExecStart=/usr/sbin/nagios -d /etc/nagios/nagios.cfg (code=exi>
   Main PID: 922 (nagios)
      Tasks: 6 (limit: 10502)
     Memory: 16.1M (peak: 16.9M)
        CPU: 106ms
   CGroup: /system.slice/nagios.service
           └─922 /usr/sbin/nagios -d /etc/nagios/nagios.cfg
             └─923 /usr/sbin/nagios --worker /var/spool/nagios/cmd/nagios.qh
               └─924 /usr/sbin/nagios --worker /var/spool/nagios/cmd/nagios.qh
                 └─925 /usr/sbin/nagios --worker /var/spool/nagios/cmd/nagios.qh
                   └─926 /usr/sbin/nagios --worker /var/spool/nagios/cmd/nagios.qh
                     └─936 /usr/sbin/nagios -d /etc/nagios/nagios.cfg

Mar 05 17:41:24 server4 nagios[922]: qh: core query handler registered
Mar 05 17:41:24 server4 nagios[922]: qh: echo service query handler registered
Mar 05 17:41:24 server4 nagios[922]: qh: help for the query handler registered
Mar 05 17:41:24 server4 nagios[922]: wproc: Successfully registered manager as >
Mar 05 17:41:24 server4 systemd[1]: Started Nagios Core 4.4.14.
lines 1-23
```

```
[jalene@server4 ~]$ rpm -qa | grep nagios
nagios-common-4.4.14-4.el9.x86_64
nagios-plugins-2.4.12-4.el9.x86_64
nagios-selinux-4.4.14-4.el9.noarch
nagios-plugins-users-2.4.12-4.el9.x86_64
nagios-plugins-swap-2.4.12-4.el9.x86_64
nagios-plugins-ssh-2.4.12-4.el9.x86_64
nagios-plugins-procs-2.4.12-4.el9.x86_64
nagios-plugins-ping-2.4.12-4.el9.x86_64
nagios-plugins-load-2.4.12-4.el9.x86_64
nagios-plugins-http-2.4.12-4.el9.x86_64
nagios-plugins-disk-2.4.12-4.el9.x86_64
nagios-4.4.14-4.el9.x86_64
[jalene@server4 ~]$
```

This also verifies the installation and service of Nagios in CentOS server. I only ran the verification on the Terminal due to difficulty of navigating and CentOS and the need to configure the username and password on CentOS to view it on the browser. However, these screenshots still verify that Nagios is successfully installed and is active.

Reflections:

Answer the following:

1. What are the benefits of having an availability monitoring tool?

An availability monitoring tool helps check if servers, systems, or network services are running properly. It quickly alerts administrators when something stops working so they can fix the problem right away. This helps reduce downtime and keeps systems more reliable. It also allows administrators to monitor the health and performance of systems, making it easier to find problems and maintain the network.

Conclusions:

This hands-on activity focuses on installing Nagios in both Ubuntu and CentOS servers. I started with searching what the dependencies are for both servers and ensured that dependencies will be installed before installing Nagios. After that, I wrote the playbook for installing Nagios on both servers and started the service automatically after installation. Finally, the main playbook contains all the roles created for both pre-requisites and Nagios installation. Verifications were also provided after running the playbook. This activity is quite difficult especially on the installation for CentOS as I am not that familiar with it, but regardless, all tasks were completed and accomplished.